Elsewhere: Devpost Writeup Context

Source material for drafting the Devpost submission. Written from the actual codebase (github.com/vlww/Elsewhere) as of Oct 29, 2026.

Instructions for the writer. Use this to draft a Devpost writeup with these sections: Inspiration, What It Does, How I Built It, Challenges, Accomplishments, What I Learned, What's Next. Write in first person singular ("I"). Keep it plain and specific: no em dashes, no hype words like "revolutionary" or "seamless", no filler. Prefer concrete examples (boba, Bob Ross) over abstractions. Only claim what is listed here; the "Accuracy guardrails" section lists what not to say.

At a glance

- **Name:** Elsewhere
- **Tagline:** "Your city, through your taste."
- **One-liner:** A map-first guide that finds places you'd actually enjoy spending time in, ranked by what you need and what you love, and turns them into routed, timed outings you can edit in plain language.
- **Live:** findelsewhere.vercel.app | **Code:** github.com/vlww/Elsewhere (MIT license)
- **Audience:** students and young adults who want somewhere new to work, study, read or hang out.
- **Built with:** Next.js 16, React 19, TypeScript, MapLibre GL, three.js, Claude (Anthropic API), Qloo Insights API, Google Places API (New), OSRM routing, OpenStreetMap / OpenFreeMap, sharp, zod, zustand, Tailwind CSS, Vercel.

The problem (for Inspiration)

- People get tired of working, studying and hanging out in the same few places, and want a change of scenery.
- They want new places they will *actually* like, not just new places.
- Asking a maps app for "food near me" or "cafe near me" ranks by convenience and distance. It doesn't know you or your taste.

Supporting statistics (use as given)

- **47%** of people who work from cafés say they do it because they enjoy the change of environment (O2 Business survey).
- **58%** of consumers are frustrated by impersonal experiences from brands that aren't relevant to them (MoEngage, 2023).
- **150%** growth in searches for local places that include "near me", faster than comparable searches (Google).

Framing: people already rely on "near me", but "near me" ranks by distance, not by who you are. Elsewhere asks what you need right now and what you love, and uses both.

What it does (features)

1. Conversational guide

- A chat panel ("Guide") where you describe what you want in your own words, e.g. "I need to lock in for two hours, but I can't stand sterile libraries" or "I want to get boba and then spend a slow afternoon somewhere green to take photos."
- The guide acts instead of interviewing you: it searches, moves the map, highlights places, saves, compares, adds interests, plans and edits trails. Results update live on the map and in the panels while it works.

2. Explore: real places, ranked two ways

- Real places nearby from Google Places (worldwide), with OpenStreetMap as a fallback and a built-in demo area (Washington Square / Greenwich Village, NYC).
- Every place gets a practical score (fit for the activity, distance, listed hours and amenities) and a taste-aware score that adds cultural affinity from Qloo. Each card has a "Why this place?" breakdown.
- Category chips filter to only the kinds you pick (cafés, bookstores, galleries, parks, tea houses, food, and more).
- Save places, compare 2 to 3 side by side, give "not for me" feedback that changes later results.

3. Your Taste

- Framed as "every aspect of your style", not a destination.
- **Vibe cards** for people who can't name their taste: Cozy, Quiet, Artsy, Vintage, Minimalist, Lively, Hidden gem, Romantic, Bohemian. Each is a real Qloo place tag, so one tap shapes recommendations.
- **Search anything you love:** films, artists, authors, brands, shows (Qloo entities) and everyday interests like specialty coffee, natural wine, live music, hiking, pickleball (Qloo tags).
- **Add it as you typed it:** if nothing matches exactly ("boba"), you can still add it. Elsewhere translates it into Qloo's vocabulary: boba becomes Bubble Tea, brutalism becomes a décor style, pickleball becomes Pickleball court, and something vague like "cozy rainy days" falls back to the closest kind of place (cafés).

4. Taste Layer (2D and 3D)

- A map toggle that lights up the places that best match your taste (the strongest 14, with the top quarter as "standouts").
- Each highlighted place glows in its own pastel colour taken from its photo (a logo, a storefront, the food); places without a usable photo get a colour from the sunset palette.
- Highlights radiate outward: real neighbouring buildings are tinted in concentric rings that fade with distance.
- Example: with Bob Ross (the painter) as an interest, 42 nearby places match, led by Washington Square Park and Jefferson Market Garden. With "Vintage", 33 match.

5. Explore and Immersive map modes

- **Explore:** a recoloured 2D vector map in a sunset pastel style, with custom pins, halos and building tints.

- **Immersive:** a 3D city view. Taste-matched places rise as lit, textured 3D buildings built from their real OpenStreetMap footprints (facade windows, cornices, awnings, rooftop details, festoon lights), with a fade-in animation and a soft aura.
- Compass that rotates with the map, zoom limits in Immersive, and a low-power toggle that reduces 3D effects.

6. Taste Trails: routed, timed outings

- Turns a request into an outing through 2 to 5 real places, with real walking (or cycling/driving) routes, travel times, arrival times and opening hours checked at the time you'd arrive.
- **What you asked for comes first:** a named stop ("boba") is found by text search and only real matches can fill it. Taste only chooses between real matches. If nothing matches, the trail says so plainly.
- Edit in plain language: "swap the last stop for something artsy", "cheaper", "more time at stop 2", "finish near where I started". Only the stops you touched change.
- One-tap presets (A creative afternoon, The slow afternoon, Before dinner, A date, ...), time budgets (1, 2, 3 hr), "Surprise me", save trails, and a playable fly-through tour.
- Sensible timing: at 1 AM, "an afternoon of..." is planned for 1 PM; with no time given in the middle of the night it plans for later in the day, unless you clearly mean tonight.
- The route renders as an animated gradient line in 2D and a glowing tube with stop beacons in 3D.

7. Honesty built in

- Nothing is invented: hours, prices, Wi-Fi and ratings appear only when a source lists them; otherwise "not listed".
- Estimates (time at each stop, arrival times) are labelled as estimates, and every trail lists its assumptions.

How I built it (technical detail)

Stack and architecture

- Next.js 16 (App Router) with React 19 and TypeScript; deployed on Vercel. About 10,500 lines of TypeScript.
- API routes: `chat` (streaming agent), `recommend`, `trail`, `entities`, `geocode`, `photo`, `place-colors`, `status`. All third-party keys stay on the server; Google photos are proxied so the key never reaches the browser.
- One zustand store is shared by the map, the panels and the agent. All tool inputs and API responses are validated with zod.

The agent (Claude)

- Claude via the Anthropic TypeScript SDK in a streaming tool-use loop, with about 20 typed tools: `search_places`, `set_location`, `find_cultural_entities`, `add_taste_signal`, `set_taste_layer`, `focus_place`, `compare_places`, `save_place`, `plan_trail`, `revise_trail`, `set_trail_story`, and more.

- Tools run the real pipeline on the server and emit typed actions (results, focus, trail, add taste...) that the client applies, so the map updates as the agent works.
- The browser sends a session snapshot each turn (location, interests, visible results, current trail), and the conversation is append-only.
- The agent is the intent parser: it turns "boba then a green afternoon" into a validated trail spec (ordered kinds of place, literal per-stop requests, time budget, mode, price cap, mood words).
- Cost control: runs on Claude Sonnet 5.5 at low effort for the demo, with prompt caching on the system prompt and the growing conversation, capped output and tool rounds. A tiny Claude Haiku call (fractions of a cent) translates free-typed interests when Qloo has no exact tag.

Qloo (cultural intelligence)

- Entity search (`/search`) for films, artists, people, books, brands; tag search (`/v2/tags`) for vibes, cuisines, activities and genres.
- `/v2/insights` with `filter.type=urn:entity:place`, the user's interests as signals, a location point and radius, category tags, and explainability, so each match says which interest it came from.
- Titles/people and general tags are queried separately and merged, because in one request the titles drowned out the tags.
- Qloo places are matched to the place inventory by name tokens and distance (under 150 m). Unmatched Qloo places join the candidate pool, and matched ones fill in missing hours, prices or photos from Qloo.
- Cleanup of noisy search: Qloo's tag search returns things like nearby businesses ("Boba Guys"), single menu items ("Boba Sundae") and rare tags ("Bob" as a dish). Elsewhere filters by tag family and popularity, requires names to fit what was typed (with one typo allowed on longer words), and dedupes singular/plural.

Places, ranking and routing

- Provider chain: Google Places (New) Nearby and Text Search, then Overpass (OpenStreetMap) if configured, then a bundled OpenStreetMap snapshot of the demo area (549 places).
- Deterministic, inspectable ranking: hard constraints applied only where data is trustworthy (unknown values never exclude a place), then a weighted score. Practical: activity fit 0.45, proximity 0.30, attributes 0.20, feedback 0.05. Taste-aware adds a cultural factor (0.32), with Qloo affinity rank-normalised within the result set.
- Opening hours: all providers are normalised to OpenStreetMap hours syntax and evaluated at the user's local time.
- Routing with OSRM (FOSSGIS servers for foot, bike and car): a duration matrix to choose the stop order, then per-leg street geometry. If routing is down, legs fall back to clearly labelled straight-line estimates.
- Trail planner: beam search over stop combinations (score minus a walkability penalty, max hop), permutation search for the fastest order with a gentle arc preference (a meal last), arrival-time hours checks that swap stops that would be closed, and time-budget fitting. Revisions lock untouched stops.

Map rendering

- MapLibre GL with OpenFreeMap vector tiles and a custom recoloured style (sunset pastels).

- Taste Layer in 2D: GeoJSON layers inserted under the building layer for heat, halos, ripple rings and tinted real building footprints queried from the vector tiles.
- Immersive 3D: a three.js custom layer inside MapLibre. Taste buildings are extruded from real OSM footprints with procedural facade textures and rooftop details; the trail is a vertex-coloured tube with an additive glow, a moving "comet", and beacon columns.
- Place colours: the server downsamples each place photo with sharp, finds the dominant saturated hue (ignoring white walls, black text and grey pavement), and softens it into a pastel fill and deeper accent. This runs server-side because Qloo's image CDN blocks browser access.

Design

- Pirata One display face, sunset gradient wordmark (orange to dark purple), and a custom logo: a sun dome over three stripes that form an "E", all inside one circle.
- Accessible markers (real buttons with labels, shapes and icons, not just colour), keyboard and Escape support, and reduced-motion support.

Challenges (real ones from the build)

- **Taste overriding the request.** Early on, "get boba, then nature shoots" produced a hookah cafe because the planner only knew "cafe" and taste picked the highest scorer. Fix: per-stop literal requests found by Google text search plus synonyms (boba is bubble tea), only real matches can fill that stop, and taste only breaks ties.
- **Noisy taste search.** Typing "boba" returned a sculptor, a TV show and menu items, but no boba. Fix: stricter tag ranking and name filtering, plus an "add as typed" path that translates the interest with Qloo, a synonym list, and a cheap Claude Haiku call, falling back to the closest kind of place.
- **Matching two datasets.** Qloo places and Google/OSM places are different records of the same venues. Matching by name tokens and distance, and handling chains and duplicates, took several iterations.
- **Making highlights look intentional.** The first "radiating" glow looked random because it was computed per map tile by edge distance. Rebuilt it from building centroids in concentric rings, owned by the nearest match. Layer ordering in MapLibre also hid the effects under buildings until they were inserted in the right place.
- **Syncing three.js with MapLibre.** The 3D layer had to stay in step with the map camera, appear after the style loaded, and fade smoothly.
- **Photo colours across origins.** The browser couldn't read Qloo-hosted images, so colour extraction moved to the server with sharp and an allowlist of image hosts.
- **Time and honesty.** Planning at 1 AM would have scheduled closed places "now". Added part-of-day and late-night logic, arrival-time hours checks, and labelled assumptions.
- **API cost.** Keeping the guide affordable with limited credits: a cheaper model, low effort, prompt caching, and caps on output and tool rounds.

Accomplishments

- A taste-aware map that works in both 2D and an immersive 3D city, with the same matches and reasons in both.

- Trails that respect what you asked for first, with real routes, times and hours, editable in plain language.
- Taste input that works for anyone: vibes for people who can't describe their taste, and "add as typed" for anything unusual.
- Every highlight is explainable and every estimate is labelled; nothing about a place is invented.
- Works worldwide with live data, and still works with no keys at all in the demo area.
- Built and deployed solo, with a custom visual identity.

What I learned

- Taste should shape choices, not override needs: literal requests first, taste second.
- Real-world data is messy. Matching records across providers and cleaning noisy search results mattered as much as the AI.
- Agents work best with a small set of typed, validated tools that drive real code paths, instead of the model inventing facts.
- People find it hard to describe their own taste; offering vibes and accepting free text works better than asking for a list of favourite films.
- Practical skills: MapLibre custom layers, three.js in a map, OSRM routing, Qloo Insights, prompt caching and cost control.

What's next (ideas, not built yet)

- Group trails that blend several people's tastes.
- Sharing trails with a link, and accounts to sync taste and saved places across devices.
- Transit routing (today transit falls back to walking routes, labelled as such).
- Events and pop-ups from Qloo and local listings, not just places.
- Learning taste over time from saves and feedback.
- A mobile app with offline city packs.

Accuracy guardrails (do not claim)

- Do not say Elsewhere trains its own ML model. Ranking is a deterministic weighted score; cultural affinity comes from Qloo.
- Do not present the overall score as a Qloo score. It is Elsewhere's estimate that includes Qloo's affinity.
- Do not claim transit directions, accounts, or cross-device sync; these are not built.
- Saved places and picked interests are stored on the device (localStorage), not in a database.
- Opening hours, prices and amenities come from the listing provider and can be missing; the app says "not listed".
- The demo area data is a real OpenStreetMap snapshot; without a Qloo key, taste signals are simulated and labelled as such.